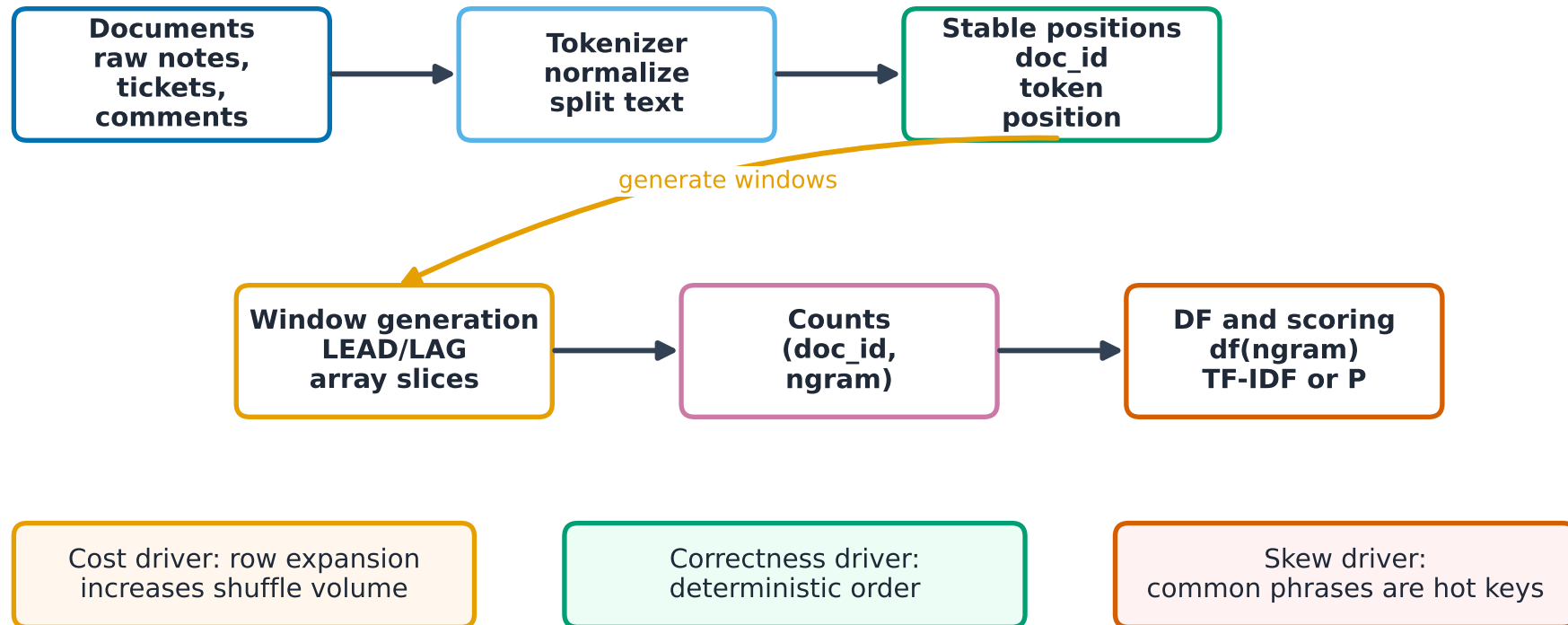


N-gram pipeline execution flow



Practical message: n-gram generation changes row counts, shuffle volume, and partition balance.